

RS-MILP-02-01: Configure ADMM(Alternating Direction Method of Multipliers) for distributed approach of MILP

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1. Generate A matrix from contact chart

```
clear;
clc;

addpath('/Library/gurobi1300/macos_universal2/matlab')
savepath

addpath ~/Desktop/Redstone_MILP/RS_MILP_01_Config_MILP/
load('E0IR_48_SATs_12_Orbit_Planes_98_inc_7days_access_interval.mat','E0IR_a
ccess_interval')

% 기준 시각 정의

T = E0IR_access_interval;

t0 = datetime(2030, 1, 1, 0, 0, 0, 'TimeZone', 'UTC');

% --- Source / Target을 string으로 통일 ---
src = string(T.Source);
tgt = string(T.Target);

% --- Ground 번호 추출 ---
gTok = regexp(src, 'Ground[_\s]*Point[_\s]*(\d+)', 'tokens', 'once');
groundNum = cellfun(@(c) str2double(c{1}), gTok);

% --- Satellite 번호 추출 ---
sTok = regexp(tgt, '(?i)SAT[^\s-0-9]*(\d+)', 'tokens', 'once');
satNum = cellfun(@(c) str2double(c{1}), sTok);

% --- Start / End → datetime ---
st = T.StartTime;
en = T.EndTime;

if ~isdatetime(st)
    st = datetime(string(st), 'InputFormat', 'dd-MMM-yyyy HH:mm:ss');
```

```

end
if ~isdatetime(en)
    en = datetime(string(en), 'InputFormat','dd-MMM-yyyy HH:mm:ss');
end

% --- 중간 시각 ---
midTime = st + (en - st)/2;

% --- 기준 시각 대비 초(second) 차이 → integer ---
timeSec = seconds(midTime - t0);
timeSec = double(round(timeSec)); % 정수화 (필요시 floor / ceil로 변경 가능)

contact_index = (1:length(st))';

% --- 최종 3-column matrix ---
A_matrix = [double(satNum), double(groundNum), timeSec, contact_index];

A_matrix = sortrows(A_matrix, 3);

A_matrix = A_matrix(1:1000,:);
Original_A = T(A_matrix(:,4),:);

N = length(A_matrix(:,1));

```

2. Extract Key parameters from A matrix

```

t = A_matrix(:,3);

% Satellite Cadence Constraint
tau = 20;

% Number of SATs
p = 48;

% Number of GSs
q = 54;

% Total number of contact
N = length(A_matrix(:,1));

% Number of contact for each SAT
S_i_vec = zeros(p,1);

for sat_index = 1:p

```

```

    S_i_vec(sat_index) = nnz(A_matrix(:,1) == sat_index);
end

% Number of contact for each GS
G_j_vec = zeros(q,1);
for gs_index = 1:q
    G_j_vec(gs_index) = nnz(A_matrix(:,2) == gs_index);
end

```

3. Selection matrix generation from given constant parameters

3.1. $E_{S_i}^1$: Selection matrix from A-matrix to each satellite's contact sequence

```

E1_Si = struct();

for i = 1:p

    E1_Si_mat = zeros(S_i_vec(i),N);

    a_i = find(A_matrix(:,1) == i);

    for j = 1:length(a_i)
        E1_Si_mat(j, a_i(j)) = 1;
    end

    E1_Si.(['sat',num2str(i)]) = E1_Si_mat;

end

```

3.2. $E_{S_{i,x}}^2, E_{S_{i,t}}^2$: Selection matrix for Δt of each satellite from $|S_i|$

```

E2_Si_x = struct();

for i = 1:p
    E2_Si_x_mat = [];
    for alpha = 1:S_i_vec(i)-1
        E2_Si_x_alpha = zeros(S_i_vec(i)-alpha, S_i_vec(i));
        E2_Si_x_alpha(:,alpha) = 1;
        for beta = alpha+1:S_i_vec(i)
            E2_Si_x_alpha(beta-alpha,beta) = 1;
        end
        E2_Si_x_mat = [E2_Si_x_mat;E2_Si_x_alpha];
    end
end

```

```

    end
    if isempty(E2_Si_x_mat)
        E2_Si_x_mat = 0;
    end
    E2_Si_x.(['sat', num2str(i)]) = E2_Si_x_mat;
end

E2_Si_t = struct();

for i = 1:p
    E2_Si_t_mat = [];
    for alpha = 1:S_i_vec(i)-1
        E2_Si_t_alpha = zeros(S_i_vec(i)-alpha, S_i_vec(i));
        E2_Si_t_alpha(:,alpha) = -1;
        for beta = alpha+1:S_i_vec(i)
            E2_Si_t_alpha(beta-alpha,beta) = 1;
        end
        E2_Si_t_mat = [E2_Si_t_mat;E2_Si_t_alpha];
    end
    if isempty(E2_Si_t_mat)
        E2_Si_t_mat = 0;
    end
    E2_Si_t.(['sat', num2str(i)]) = E2_Si_t_mat;
end

```

3.3. $E_{G_j}^1$: Selection matrix from A-matrix to each Ground Point's revisit sequence

```

E1_Gj = struct();

for j = 1:q
    E1_Gj_mat = zeros(G_j_vec(j)+2, N+2);

    E1_Gj_mat(1,1) = 1;
    b_j = find(A_matrix(:,2) == j);

    for alpha = 1:length(b_j)
        E1_Gj_mat(alpha+1, b_j(alpha)+1) = 1;
    end
    E1_Gj_mat(G_j_vec(j)+2, N+2) = 1;
    E1_Gj.(['gs', num2str(j)]) = E1_Gj_mat;
end

```

3.4. $E_{G_{j,x}}^2, E_{G_{j,t}}^2$: Selection matrix for Δt of each GS from $|G_j|$

```

E2_Gj_x = struct();

for j = 1:q
    E2_Gj_x_mat = [];
    for alpha = 1:G_j_vec(j)+1
        E2_Gj_x_alpha = zeros(G_j_vec(j)+2-alpha, G_j_vec(j)+2);
        E2_Gj_x_alpha(:,alpha) = 1;
        for beta = alpha+1:G_j_vec(j)+2
            if alpha + 1 <= beta-1
                E2_Gj_x_alpha(beta-alpha, alpha+1:beta-1) = -1;
            end
            E2_Gj_x_alpha(beta-alpha,beta) = 1;
        end
        E2_Gj_x_mat = [E2_Gj_x_mat;E2_Gj_x_alpha];
    end
    if isempty(E2_Gj_x_mat)
        E2_Gj_x_mat = 0;
    end
    E2_Gj_x.(['gs',num2str(j)]) = E2_Gj_x_mat;
end

E2_Gj_t = struct();

for j = 1:q
    E2_Gj_t_mat = [];
    for alpha = 1:G_j_vec(j)+1
        E2_Gj_t_alpha = zeros(G_j_vec(j)+2-alpha, G_j_vec(j)+2);
        E2_Gj_t_alpha(:,alpha) = -1;
        for beta = alpha+1:G_j_vec(j)+2
            E2_Gj_t_alpha(beta-alpha,beta) = 1;
        end
        E2_Gj_t_mat = [E2_Gj_t_mat;E2_Gj_t_alpha];
    end
    if isempty(E2_Gj_t_mat)
        E2_Gj_t_mat = 0;
    end
    E2_Gj_t.(['gs',num2str(j)]) = E2_Gj_t_mat;
end

```

4. Derivation of A, b, C, d, E, f, G revisit time problem to MILP

4.1 A, b for L_1 problem , A_{s_i}, b_{s_i} for BCD/ADMM , P – matrix for Permutation, U, V for sizing index

```

A = [];
b_mat = [];
P_matrix_transpose = [];

```

```

U = 0;
V = 0;

for i = 1:p
    if isempty(E1_Si.(['sat', num2str(i)]))
        continue;
    end

    A_i = E2_Si_x.(['sat', num2str(i)]) * E1_Si.(['sat', num2str(i)]);
    A = [A;A_i];
    b_i_mat = E2_Si_t.(['sat', num2str(i)]) * E1_Si.(['sat', num2str(i)]);
    b_mat = [b_mat; b_i_mat];

    A_si_info.(['sat', num2str(i)]) = E2_Si_x.(['sat', num2str(i)]);
    b_si_info.(['sat', num2str(i)]) = ones(length(b_i_mat(:,1)),1) + 1/tau *
b_i_mat* A_matrix(:,3);

    P_matrix_transpose = [P_matrix_transpose; E1_Si.(['sat', num2str(i)])];

    E1_si = E1_Si.(['sat', num2str(i)]);
    S_i_info.(['sat', num2str(i)]) = length(E1_si(:,1));

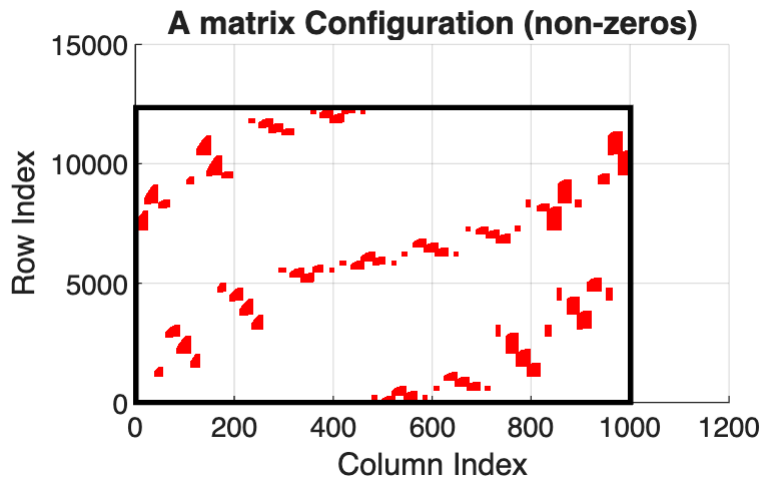
    U_i_info.(['sat', num2str(i)]) = U;
    V_i_info.(['sat', num2str(i)]) = N - length(E1_si(:,1)) - U;
    U = U + length(E1_si(:,1));

end

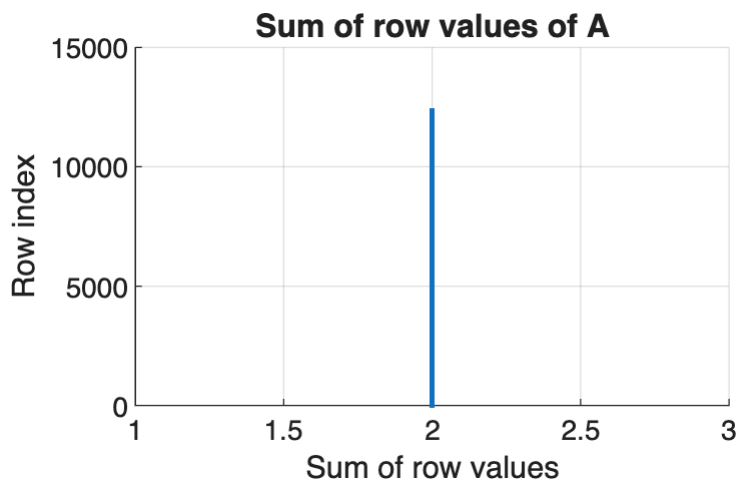
b = ones(length(b_mat(:,1)),1) + 1/tau * b_mat * A_matrix(:,3);
P_matrix = P_matrix_transpose';

[A_row, A_col] = find(A==1);
figure;
hold on
scatter(A_col,A_row,'r','.')
rectangle('Position',[1 1 max(A_col), max(A_row)], ...
    'EdgeColor','k', ...
    'LineWidth',2);
title('A matrix Configuration (non-
zeros)','FontSize',12,'FontWeight','bold')
xlabel('Column Index')
ylabel('Row Index')
grid on

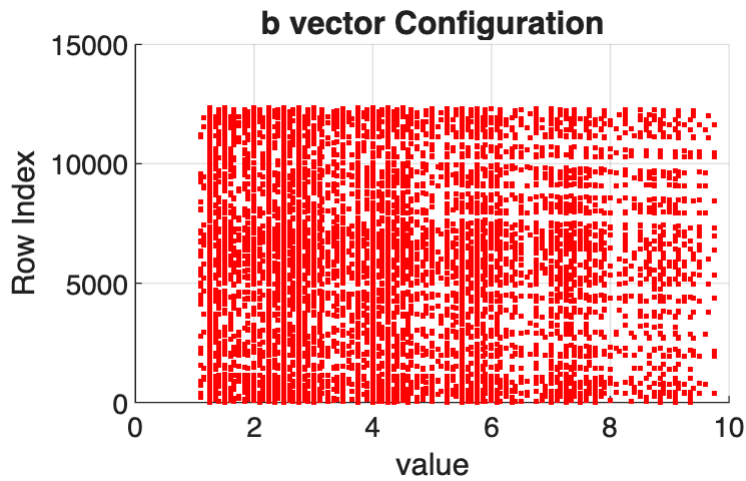
```



```
row_sum = sum(A, 2);           % 각 row의 합 (N×1)
figure;
scatter(row_sum,1:length(row_sum),'.'); % x축은 자동으로 row index (1:N)
grid on
ylabel('Row index')
xlabel('Sum of row values')
title('Sum of row values of A','FontSize',12,'FontWeight','bold')
```



```
figure;
scatter(b,1:length(b),'r','.');
title('b vector Configuration','FontSize',12,'FontWeight','bold')
xlabel('value')
ylabel('Row Index')
xlim([0,10])
grid on
```



4.2 C, d for L_∞ problem

```

C = [];
d = [];

t = A_matrix(:,3); % (N×1) time vector in seconds (already
prepared)
t_aug = [min(A_matrix(:,3)); t; max(A_matrix(:,3))+1]; % (N+2 × 1)

for j = 1:q
    key = ['gs', num2str(j)];

    if isempty(E1_Gj.(key))
        continue;
    end

    % --- shorthand ---
    E1 = E1_Gj.(key); % selection matrix for this ground
    E2x = E2_Gj_x.(key);
    E2t = E2_Gj_t.(key);

    % =====
    % Build C_j (constant)
    % =====
    % D_j = diag( E2t*E1 * t_aug ) (vector -> diagonal matrix)
    Dj_vec = (E2t * E1) * t_aug; % (#rows_j × 1)
    Dj = spdiags(Dj_vec, 0, length(Dj_vec), length(Dj_vec)); % sparse diag
    is safer

    % P = [0_{1×N}; I_{N×N}; 0_{1×N}] -> (N+2 × N)
    N = length(t);
    P = [zeros(1,N); speye(N); zeros(1,N)];

```



```

% C_j = D_j * (E2x*E1) * P
Cj = Dj * (E2x * E1) * P;

% =====
% Build d_j (constant)
% =====
% v = [1; 0_{N×1}; 1] - 1 = [0; -1...; 0]
v = [1; zeros(N,1); 1]; % (N+2 × 1)

% d_j = D_j * (E2x*E1) * v
dj = Dj * ((E2x * E1) * v - ones(length(Dj(:,1)),1));

% --- stack ---
C = [C; Cj];
d = [d; dj];
end

% [C_row, C_col] = find(C~=0);
%
% figure;
% scatter(C_row, C_col,'r','.')
% grid on

```

4.3 E, f, G for L_2 Optimization problem

```

% =====
% Build E, f, G (2nd image)
% =====

E_ori = [];
F = [];

t = A_matrix(:,3); % (N×1)
N = length(t);

t_aug = [min(t); t; max(t)+1]; % (N+2 × 1), same as your code

% P = [0_{1×N}; I_N; 0_{1×N}] → (N+2 × N)
P = [zeros(1,N); speye(N); zeros(1,N)];

% this is the (N+2 × 1) vector [1; 0_N; 1]
one_aug = [1; zeros(N,1); 1];

% For G, we will build the stacked matrix Bt = [E2t*E1; ...] first
(optional),
% or directly build gvec = Bt*t_aug.

```

```

gvec = [];    % = Bt * t_aug (will be stacked across j)

for j = 1:q
    key = ['gs', num2str(j)];

    if isempty(E1_Gj.(key))
        continue;
    end

    % --- shorthand ---
    E1 = E1_Gj.(key);          % selection matrix for this ground
    E2x = E2_Gj_x.(key);
    E2t = E2_Gj_t.(key);

    % Common blocks
    Sx = (E2x * E1);           % (#rows_j × (N+2))
    St = (E2t * E1);           % (#rows_j × (N+2))

    % -----
    % E_j = Sx * P (constant)
    % -----
    Ej = Sx;                   % (#rows_j × N)
    E_ori = [E_ori; Ej];

    % -----
    % f_j = Sx]
    % -----
    Fj = Sx;                   % (#rows_j × 1)
    F = [F; Fj];

    % -----
    % gvec_j = St * t_aug
    % -----
    gvec_j = St * t_aug;       % (#rows_j × 1)
    gvec = [gvec; gvec_j];
end

% Dj = spdiags(Dj_vec, 0, length(Dj_vec), length(Dj_vec));

E = E_ori * P;
f_vector = F * one_aug;

% -----
% G = gvec * gvec'
% -----
% WARNING: this can be very large/dense if gvec is long.
% If you only need y'*G*y, note that y'*(gvec*gvec')*y = (gvec'*y)^2.
G = spdiags(gvec, 0, length(gvec), length(gvec)); % (M×M)

```

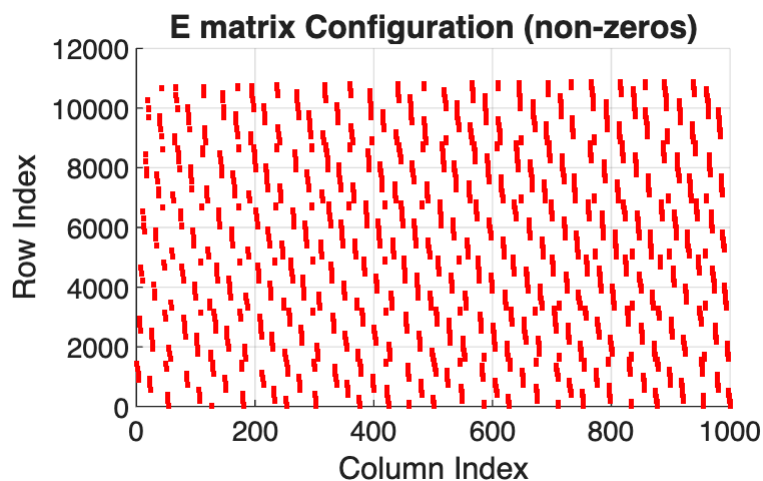
```

[E_one_col, E_one_row] = find(E == 1);
[E_minus_one_col, E_minus_one_row] = find(E == -1);

figure;
hold on
scatter(E_minus_one_row, E_minus_one_col, 'blue', '.')
scatter(E_one_row, E_one_col, 'red', '.')

grid on
title('E matrix Configuration (non-
zeros)', 'FontSize', 12, 'FontWeight', 'bold')
xlabel('Column Index')
ylabel('Row Index')
grid on

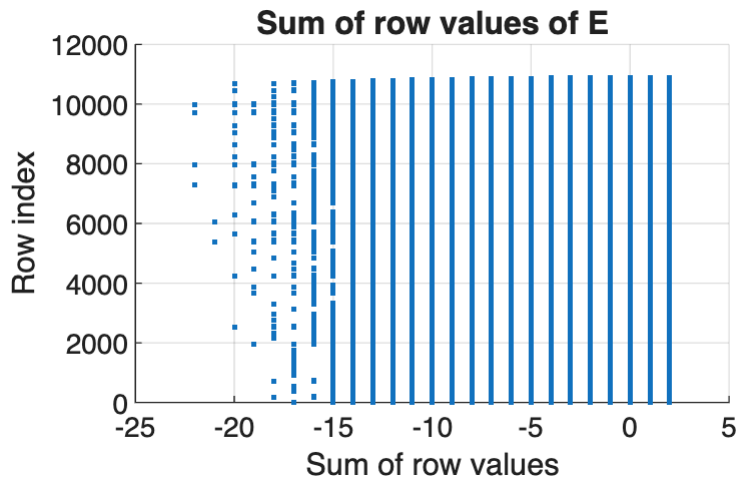
```



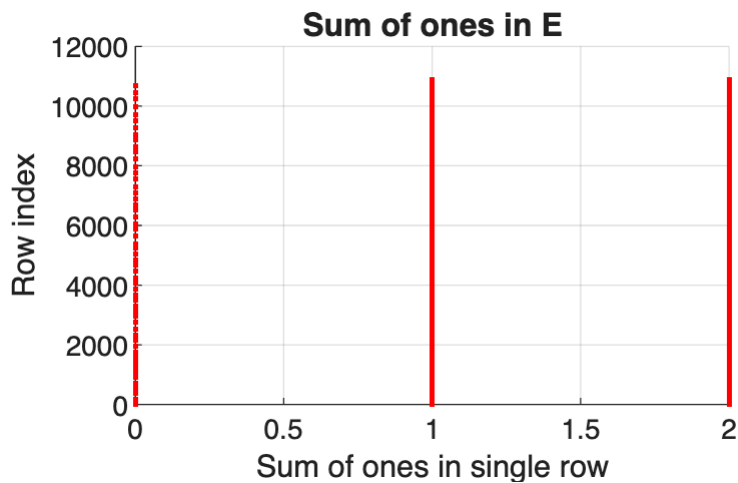
```

row_sum = sum(E, 2);           % 각 row의 합 (N×1)
figure;
scatter(row_sum, 1:length(row_sum), '.'); % x축은 자동으로 row index (1:N)
grid on
ylabel('Row index')
xlabel('Sum of row values')
title('Sum of row values of E', 'FontSize', 12, 'FontWeight', 'bold')

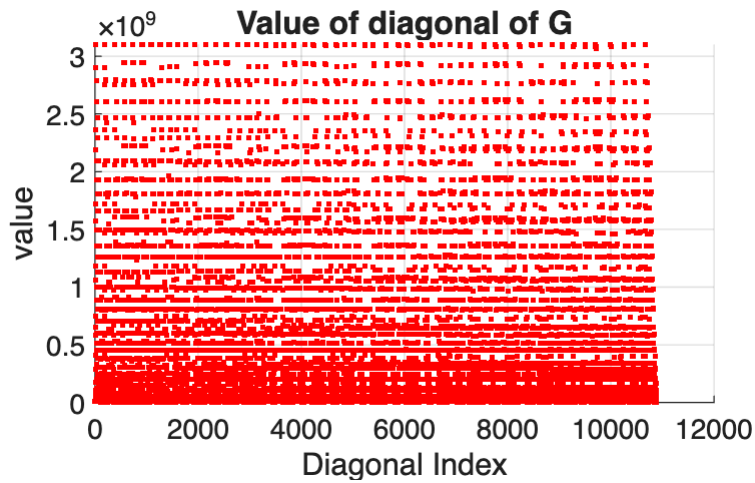
```



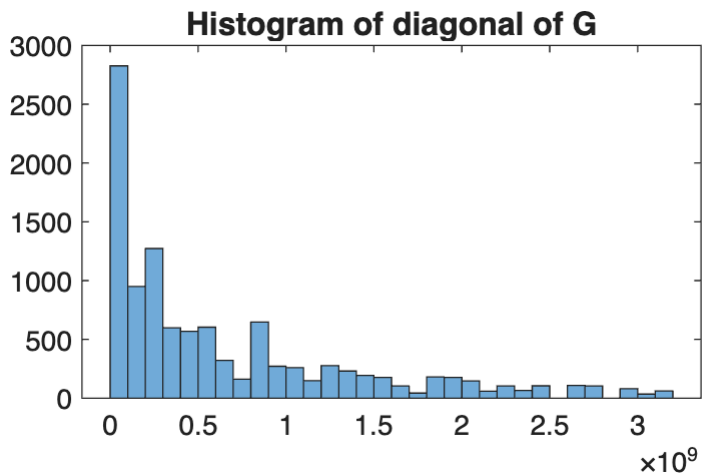
```
row_one_sum = sum(E == 1, 2);           % 각 row의 합 (N×1)
figure;
scatter(row_one_sum,1:length(row_one_sum),'.','r');           % x축은 자동으로
row index (1:N)
grid on
ylabel('Row index')
xlabel('Sum of ones in single row')
title('Sum of ones in E','FontSize',12,'FontWeight','bold')
```



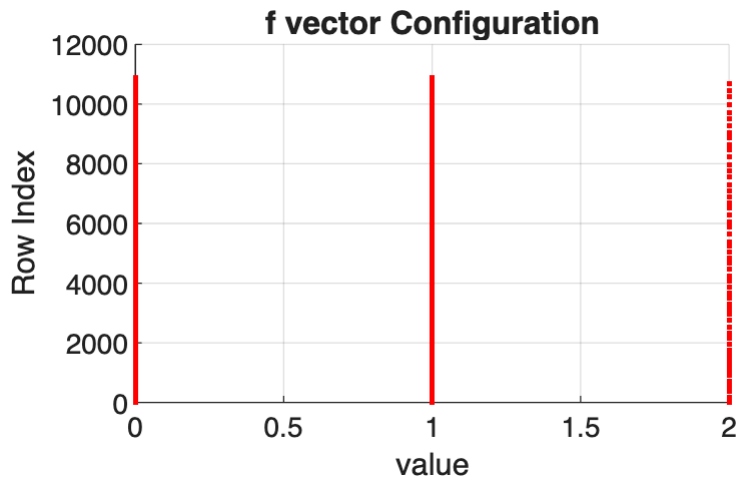
```
scatter(1:length(gvec),gvec.^2,'r','.')
title('Value of diagonal of G','FontSize',12,'FontWeight','bold');
ylabel('value')
xlabel('Diagonal Index')
grid on
```



```
figure;
histogram(gvec.^2)
title('Histogram of diagonal of G','FontSize',12,'FontWeight','bold');
```



```
scatter(f_vector,1:length(f_vector),'r','.')
title('f vector Configuration','FontSize',12,'FontWeight','bold')
xlabel('value')
ylabel('Row Index')
grid on
```



5. Solving MILP by ADMM

5.1 ADMM

```
%% =====
% Consensus ADMM for (MIQP/MILP-like) with  $z \geq E \cdot P \cdot \bar{x} + f - 1$ 
% - each agent  $i$  updates  $(x_{\{s_i\}}, z_{\{s_i\}})$  in parallel/serial
% - global  $z$  enforces consensus:  $z_{\{s_i\}} = z$ 
% - dual variables  $\lambda_i$  (same dimension as  $z$ )
%% =====

% ---- user provided initializations (keep) ----
x = zeros(length(A_matrix(:,1)),1); % global x (binary)
z = zeros(length(f_vector),1); % global consensus z (continuous)

% ---- ADMM options ----
maxIters = 200;
rho = 6 * 10e6; % penalty parameter
tol_primal = 1; % ||r|| tolerance
tol_dual = 1; % ||s|| tolerance
verbose = 1;

% ---- precompute constant matrices ----
EP = E * P_matrix; % used for z lower bound

% ---- build block index map (idx_i for each agent) ----
idx_block = cell(p,1);
for i = 1:p
    if isempty(E1_Si.(['sat', num2str(i)]))
        idx_block{i} = [];
        continue;
    end
end
```

```

    U = U_i_info.(['sat', num2str(i)]);
    S_i = S_i_info.(['sat', num2str(i)]);
    nSi = S_i;
    idx_block{i} = (U+1):(U+nSi);
end

% ---- dual variables (one lambda per agent, dimension = length(z)) ----
m = length(f_vector);
lambda = cell(p,1);
for i = 1:p
    lambda{i} = zeros(m,1);
end

% ---- histories (optional) ----
x_hist = [];
z_hist = [];
r_hist = [];
s_hist = [];

% ---- weight for quadratic on z_{s_i} ----
g = (gvec(:)).^2; % objective uses z' diag(g) z

% ---- main ADMM loop ----
z_prev = z;

for k = 1:maxIters

    % snapshot
    x_k = x;
    z_k = z;

    % stacked xbar = P' x
    xbar_k = P_matrix' * x_k;

    % store local solutions
    z_si_all = cell(p,1);
    xbar_new_all = cell(p,1);

    % ===== (1) Local updates: for each agent i, solve MIQP in (x_si,
    z_si)
    for i = 1:p
        if isempty(E1_Si.(['sat', num2str(i)]))
            z_si_all{i} = z_k; % dummy
            xbar_new_all{i} = xbar_k; % dummy
            continue;
        end

        % local constraints
        A_si = A_si_info.(['sat', num2str(i)]);
        b_si = b_si_info.(['sat', num2str(i)]);
    end
end

```

```

% current local block (from global x)
x_si_k = E1_Si.(['sat', num2str(i)]) * x_k;

% coupling base term: base0 = EP*xbar_k + f - 1
base0 = EP * xbar_k + f_vector - 1;

% EP_i: columns corresponding to agent i block inside xbar
U = U_i_info.(['sat', num2str(i)]);
S = S_i_info.(['sat', num2str(i)]);
EP_i = EP(:, (U+1):(U+S)); % m x |S_i|

% solve local MIQP:
% min_{x_si in {0,1}^n, z in R^m}
%      g' z + lambda' (z - z_k) + (rho/2) ||z - z_k||_2^2
% s.t. A_si x_si <= b_si, x_si binary
%      z_si >= base0 + EP_i*(x_si - x_si_k)
%      z_si >= 0
params = struct();
params.OutputFlag = 0;

sol = solve_local_admm_milp_gurobi(A_si, b_si, EP_i, base0, x_si_k,
...
                                g, z_k, lambda{i}, rho, params);

if ~sol.success
    if verbose
        fprintf('[ADMM] iter=%d sat%d local solve failed
(status=%s). Keep.\n', k, i, sol.status);
    end
    z_si_all{i} = z_k;
    xbar_new_all{i} = xbar_k;
    continue;
end

x_si_new = sol.x_si;
z_si_new = sol.z_si;
z_si_all{i} = z_si_new;
xbar_new_all{i} = x_si_new;
end

% ===== (2) Global x update
% 보통 consensus ADMM에서는 x를 "각 agent의 x_si를 합쳐" 업데이트해야 합니다.
% 여기서는 가장 단순히 "마지막 sweep의 xbar_new_all{i}"를 사용하거나,
% 혹은 다수결/평균 후 rounding 등을 설계할 수 있습니다.
%
% 아래는 "sequential sweep"처럼 i=1..p 순서로 반영한 결과를 사용:
xbar_seq = xbar_k;
for i = 1:p
    if isempty(idx_block{i}), continue; end

```



```

    U = U_i_info(['sat', num2str(i)]);
    S = S_i_info(['sat', num2str(i)]);
    xbar_seq(U+1:U+S) = xbar_new_all{i};
end
x = P_matrix * xbar_seq;

% ===== (3) Global z update (Option i): element-wise max of local
z_si
m = length(f_vector);
z_new = -inf(m,1);

cnt = 0;
for i = 1:p
    if isempty(E1_Si(['sat', num2str(i)])), continue; end
    z_new = max(z_new, z_si_all{i}); % element-wise max
    cnt = cnt + 1;
end

if cnt == 0
    z_new = z_k;
end

% (강력 권장) 전역 coupling 하한도 같이 만족시키기:  $z \geq EP \cdot \bar{x} + f - 1$ ,  $z \geq 0$ 
xbar = P_matrix' * x; % x는 (2) x-update 이후의 최신 값
lb1 = EP * xbar + f_vector - 1;
lb2 = zeros(m,1);
z = max(z_new, max(lb1, lb2));

% ===== (4) Dual update
for i = 1:p
    if isempty(E1_Si(['sat', num2str(i)])), continue; end
    lambda{i} = lambda{i} + rho*( z_si_all{i} - z );
end

% ===== (5) Residual checks
% primal residual:  $r_i = z_{si} - z$ 
r_norm = 0;
cnt = 0;
for i = 1:p
    if isempty(E1_Si(['sat', num2str(i)])), continue; end
    r_norm = r_norm + norm(z_si_all{i} - z, 2)^2;
    cnt = cnt + 1;
end
r_norm = sqrt(r_norm);

% dual residual:  $s = \rho \cdot (z - z_{prev})$ 
s_norm = norm(z - z_prev, 2);

% histories

```

```

x_hist = [x_hist, x];
z_hist = [z_hist, z];
r_hist = [r_hist, r_norm];
s_hist = [s_hist, s_norm];

if verbose
    fprintf('[ADMM] iter=%3d |r|=%.3e |s|=%.3e\n', k, r_norm, s_norm);
end

if (r_norm <= tol_primal) && (s_norm <= tol_dual)
    if verbose, fprintf('[ADMM] Converged at iter=%d\n', k); end
    break;
end

z_prev = z;
end

```

```

[ADMM] iter= 1 |r|=2.181e+02 |s|=3.243e+01
[ADMM] iter= 2 |r|=1.758e+02 |s|=3.403e+01
[ADMM] iter= 3 |r|=1.679e+02 |s|=2.927e+01
[ADMM] iter= 4 |r|=1.469e+02 |s|=2.592e+01
[ADMM] iter= 5 |r|=1.379e+02 |s|=2.479e+01
[ADMM] iter= 6 |r|=1.349e+02 |s|=2.386e+01
[ADMM] iter= 7 |r|=1.189e+02 |s|=2.261e+01
[ADMM] iter= 8 |r|=1.158e+02 |s|=2.131e+01
[ADMM] iter= 9 |r|=1.059e+02 |s|=2.049e+01
[ADMM] iter= 10 |r|=1.056e+02 |s|=1.955e+01
[ADMM] iter= 11 |r|=9.999e+01 |s|=1.877e+01
[ADMM] iter= 12 |r|=9.859e+01 |s|=1.843e+01
[ADMM] iter= 13 |r|=9.377e+01 |s|=1.767e+01
[ADMM] iter= 14 |r|=9.131e+01 |s|=1.740e+01
[ADMM] iter= 15 |r|=9.133e+01 |s|=1.734e+01
[ADMM] iter= 16 |r|=9.128e+01 |s|=1.690e+01
[ADMM] iter= 17 |r|=8.434e+01 |s|=1.627e+01
[ADMM] iter= 18 |r|=8.107e+01 |s|=1.543e+01
[ADMM] iter= 19 |r|=7.609e+01 |s|=1.436e+01
[ADMM] iter= 20 |r|=6.542e+01 |s|=1.311e+01
[ADMM] iter= 21 |r|=6.223e+01 |s|=1.165e+01
[ADMM] iter= 22 |r|=5.595e+01 |s|=1.112e+01
[ADMM] iter= 23 |r|=5.498e+01 |s|=1.015e+01
[ADMM] iter= 24 |r|=4.886e+01 |s|=9.775e+00
[ADMM] iter= 25 |r|=5.013e+01 |s|=9.003e+00
[ADMM] iter= 26 |r|=4.340e+01 |s|=8.806e+00
[ADMM] iter= 27 |r|=4.258e+01 |s|=7.924e+00
[ADMM] iter= 28 |r|=3.756e+01 |s|=7.674e+00
[ADMM] iter= 29 |r|=4.196e+01 |s|=7.396e+00
[ADMM] iter= 30 |r|=3.466e+01 |s|=6.888e+00
[ADMM] iter= 31 |r|=3.198e+01 |s|=5.772e+00
[ADMM] iter= 32 |r|=2.593e+01 |s|=5.352e+00
[ADMM] iter= 33 |r|=2.436e+01 |s|=4.458e+00
[ADMM] iter= 34 |r|=1.876e+01 |s|=3.981e+00
[ADMM] iter= 35 |r|=2.217e+01 |s|=3.742e+00
[ADMM] iter= 36 |r|=1.465e+01 |s|=2.839e+00
[ADMM] iter= 37 |r|=1.066e+01 |s|=1.768e+00
[ADMM] iter= 38 |r|=1.036e+01 |s|=1.889e+00
[ADMM] iter= 39 |r|=1.970e+01 |s|=2.249e+00
[ADMM] iter= 40 |r|=1.839e+01 |s|=2.659e+00
[ADMM] iter= 41 |r|=1.841e+01 |s|=3.081e+00
[ADMM] iter= 42 |r|=2.351e+01 |s|=3.065e+00
[ADMM] iter= 43 |r|=1.789e+01 |s|=2.377e+00

```

```

[ADMM] iter= 44 |r|=2.003e+01 |s|=2.050e+00
[ADMM] iter= 45 |r|=2.777e+01 |s|=3.621e+00
[ADMM] iter= 46 |r|=1.556e+01 |s|=3.310e+00
[ADMM] iter= 47 |r|=1.611e+01 |s|=1.778e+00
[ADMM] iter= 48 |r|=1.609e+01 |s|=2.400e+00
[ADMM] iter= 49 |r|=2.612e+01 |s|=3.300e+00
[ADMM] iter= 50 |r|=1.145e+01 |s|=3.010e+00
[ADMM] iter= 51 |r|=2.055e+01 |s|=2.831e+00
[ADMM] iter= 52 |r|=8.594e+00 |s|=2.757e+00
[ADMM] iter= 53 |r|=8.793e+00 |s|=1.111e+00
[ADMM] iter= 54 |r|=5.492e+00 |s|=1.000e+00
[ADMM] iter= 55 |r|=7.474e+00 |s|=1.129e-02
[ADMM] iter= 56 |r|=3.392e+00 |s|=5.643e-03
[ADMM] iter= 57 |r|=2.072e+00 |s|=1.000e+00
[ADMM] iter= 58 |r|=2.012e+00 |s|=1.411e-03
[ADMM] iter= 59 |r|=2.003e+00 |s|=8.504e-04
[ADMM] iter= 60 |r|=1.994e+00 |s|=4.252e-04
[ADMM] iter= 61 |r|=1.616e+00 |s|=2.126e-04
[ADMM] iter= 62 |r|=1.470e+00 |s|=1.063e-04
[ADMM] iter= 63 |r|=1.431e+00 |s|=5.315e-05
[ADMM] iter= 64 |r|=1.206e+00 |s|=2.657e-05
[ADMM] iter= 65 |r|=6.099e-01 |s|=1.329e-05
[ADMM] Converged at iter=65

```

```

x_opt_L2 = x_hist(:,end); % binary
x_opt_L2(abs(x_opt_L2) < 1e-1) = 0;

row_index_L2 = x_opt_L2 .* A_matrix(:,4);
row_index_L2 = row_index_L2(row_index_L2 ~= 0);

Original_A_L2 = T(row_index_L2,:);
A_matrix_L2 = A_matrix(x_opt_L2 == 1,:);

```

```

function sol = solve_local_admm_milp_gurobi(A_si, b_si, EP_i, base0,
x_si_k, ...
gvec, z_k, lambda_i,
rho, params)
% Solve local MIQP:
%
% min_{x_si in {0,1}^n, z in R^m}
% g' z + lambda' (z - z_k) + (rho/2) ||z - z_k||_2^2
% s.t.

```

```

%      A_si x_si <= b_si
%      z >= base0 + EP_i*(x_si - x_si_k)
%      z >= 0
%
% Variables v = [x; z].

if nargin < 10 || isempty(params), params = struct(); end
if ~isfield(params, 'OutputFlag'), params.OutputFlag = 0; end
if ~isfield(params, 'LogToConsole'), params.LogToConsole = 0; end
if ~isfield(params, 'LogFile'), params.LogFile = ''; end

gvec      = gvec(:);
z_k       = z_k(:);
lambda_i  = lambda_i(:);

n = size(A_si,2);      % |x_si|
m = length(base0);     % |z|

% -----
% Objective expansion:
% g'z + lambda'(z - z_k) + (rho/2)||z - z_k||^2
%
% = (rho/2) z'z + (g + lambda - rho z_k)' z + const
%
% Gurobi form: min (1/2) v'Qv + c'v
% -> put Qzz = rho*I (because (1/2) z'(rho I) z = (rho/2) z'z)
% -> linear term on z: c_z = g + lambda - rho z_k
% -----
Q = sparse(n+m, n+m);
Q(n+1:n+m, n+1:n+m) = rho * speye(m);

c = zeros(n+m,1);
c(n+1:n+m) = gvec + lambda_i - rho*z_k;

% -----
% Constraints:
% (1) A_si x <= b_si
% (2) z >= base0 + EP_i*(x - x_k)
%     -> -z + EP_i*x <= -(base0 - EP_i*x_k)
% (3) z >= 0 handled by lb on z
% -----
A1  = [A_si, sparse(size(A_si,1), m)];
rhs1 = b_si;

A2  = [EP_i, -speye(m)];
rhs2 = -(base0 - EP_i*x_si_k);

A    = sparse([A1; A2]);
rhs  = [rhs1; rhs2];
sense = repmat('<', size(A,1), 1);

```

```

% bounds
lb = -inf(n+m,1);
ub = inf(n+m,1);

lb(1:n) = 0;    ub(1:n) = 1;    % x binary bounds
lb(n+1:n+m) = 0;    % z >= 0

vtype = repmat('C', n+m, 1);
vtype(1:n) = 'B';

model.Q = Q;
model.obj = c;
model.A = A;
model.rhs = rhs;
model.sense = sense;
model.lb = lb;
model.ub = ub;
model.vtype = vtype;
model.modelsense = 'min';

% Solve
try
    result = gurobi(model, params);
catch ME
    sol.success = false;
    sol.status = ['error: ' ME.message];
    sol.x_si = [];
    sol.z_si = [];
    return;
end

sol.status = result.status;

if isfield(result, 'x') && strcmpi(result.status, 'OPTIMAL')
    v = result.x;
    sol.x_si = v(1:n);
    sol.z_si = v(n+1:n+m);
    sol.success = true;
else
    sol.success = false;
    sol.x_si = [];
    sol.z_si = [];
end
end

```

```

A_matrix_result = A_matrix;

active_satellites = unique(A_matrix_result(:,1));
cadence_matrix_original = zeros(length(active_satellites),4);

    for index = 1:length(active_satellites)
        sat_index = active_satellites(index);
        % satellite_cadence_info.(['sat',num2str(sat_index)]) =
A_matrix_result(A_matrix_result(:,1) == sat_index,:);
        satellite_cadence = A_matrix_result(A_matrix_result(:,1) ==
sat_index,:);
        cadence_info = zeros(length(satellite_cadence(:,1))-1,1);
        for i = 1:length(cadence_info(:,1))
            cadence_info(i) = satellite_cadence(i+1,3) -
satellite_cadence(i,3);
        end

        if isscalar(satellite_cadence(:,1))
            continue;
        end

        cadence_matrix_original(index,:) = [sat_index, max(cadence_info),
min(cadence_info), mean(cadence_info)];
    end

A_matrix_result = A_matrix_L2;

active_satellites = unique(A_matrix_result(:,1));
cadence_matrix_L2 = zeros(length(active_satellites),4);

    for index = 1:length(active_satellites)
        sat_index = active_satellites(index);
        % satellite_cadence_info.(['sat',num2str(sat_index)]) =
A_matrix_result(A_matrix_result(:,1) == sat_index,:);
        satellite_cadence = A_matrix_result(A_matrix_result(:,1) ==
sat_index,:);

        if isscalar(satellite_cadence(:,1))
            continue;
        end

        cadence_info = zeros(length(satellite_cadence(:,1))-1,1);

```

```

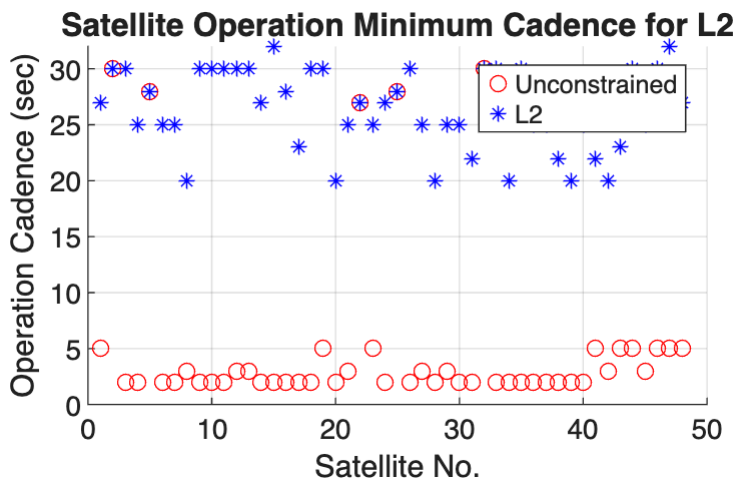
for i = 1:length(cadence_info(:,1))
    cadence_info(i) = satelllite_cadence(i+1,3) -
satelllite_cadence(i,3);
end

    cadence_matrix_L2(index,:) = [sat_index, max(cadence_info),
min(cadence_info), mean(cadence_info)];
end

cadence_matrix_L2 = cadence_matrix_L2(cadence_matrix_L2(:,3)~=0,:);

figure;
hold on
scatter(cadence_matrix_original(:,1),cadence_matrix_original(:,3),'o','red')
scatter(cadence_matrix_L2(:,1),cadence_matrix_L2(:,3),'*','blue')
title('Satellite Operation Minimum Cadence for
L2','FontSize',12,'FontWeight','bold')
legend('Unconstrained','L2')
xlabel('Satellite No.')
ylabel('Operation Cadence (sec)')
grid on

```



```

% Extract Ground Point Tables
T = Original_A_L2;

% Source 열에서 고유 값 추출
sources = unique(T.Source);

% Revisit Time Matrix (Min / Max / Mean) 생성
Revisit_Matrix = zeros(length(sources),3);
revisit_time_vector_combined = [];

t0 = datetime(2030, 1, 1, 0, 0, 0,'TimeZone','UTC');

```

```

t_start = t0 + seconds(min(A_matrix(:,3)));
t_end = t0 + seconds(max(A_matrix(:,3))+1);

% 각각의 Source 별로 테이블을 분리하여 저장
for i = 1:length(sources)
    src = sources{i};

    % Source 값이 같은 행들만 추출
    subTable = T(strcmp(T.Source, src), :);
    subTable_sorted = sortrows(subTable,4,"ascend");

    % 동적으로 변수 생성 (예: Ground_Point_1_Table)
    varName = sprintf('%s_Table', src);
    % assignin('base', varName, subTable_sorted);
    contact_tables.(sprintf('%s_Table', src)) = subTable_sorted;

    % 기존 row에 start 및 endtime 추가
    row0 = subTable_sorted(1,:);

    % -----
    % 윗줄 (t_start)
    % -----
    row_top = row0;
    row_top.IntervalNumber = NaN;      % 필요 없으면 NaN
    row_top.StartTime = t_start;
    row_top.EndTime = t_start;
    row_top.Duration = 0;
    row_top.StartOrbit = 0;
    row_top.EndOrbit = 0;

    % -----
    % 아랫줄 (t_end)
    % -----
    row_bottom = row0;
    row_bottom.IntervalNumber = NaN;
    row_bottom.StartTime = t_end;
    row_bottom.EndTime = t_end;
    row_bottom.Duration = 0;
    row_bottom.StartOrbit = 0;
    row_bottom.EndOrbit = 0;

    % -----
    % 위 + 기존 + 아래 결합
    % -----
    subTable_sorted = [row_top; subTable_sorted; row_bottom];

    % Initialize Revisit Time Vector
    revisit_time_vector = zeros(height(subTable_sorted)-1,1);

```



```

        for revisit_time_index = 1:length(revisit_time_vector)
            revisit_time_vector(revisit_time_index)
= seconds(table2array(subTable_sorted(revisit_time_index+1,4))-
table2array(subTable_sorted(revisit_time_index,5)));

            if revisit_time_vector(revisit_time_index) < 0
                revisit_time_vector(revisit_time_index) = 0;
            end
            revisit_vectors.(sprintf('%s_revisit_time_vec', src)) =
revisit_time_vector;
        end

        % 생성된 Revisit Time Vector의 Min / Max / Mean 값을 Revisit Time
Matrix에 저장
        Revisit_Matrix(i,1) = min(revisit_time_vector);
        Revisit_Matrix(i,2) = max(revisit_time_vector);

        % 재방문 주기의 평균을 구할때는 재방문 주기가 0인 데이터 포인트를 모두 제외하였음
        revisit_time_vector = revisit_time_vector(revisit_time_vector~=0);

        Revisit_Matrix(i,3) = mean(revisit_time_vector);
        revisit_time_vector_combined =
[revisit_time_vector_combined;revisit_time_vector];
        revisit_time_vector_info.(['source' num2str(i)]) =
revisit_time_vector;
    end

    % Revisit Time Matrix의 값을 그래프로 출력
    x = 1:length(Revisit_Matrix(:,1));
    min_value = (Revisit_Matrix(:,1))/3600;
    max_value = (Revisit_Matrix(:,2))/3600;
    mean_value = (Revisit_Matrix(:,3))/3600;
    output_data = [mean_value, min_value, max_value];

    figure;
    errorbar(x, mean_value, mean_value-min_value, max_value-
mean_value, '*', 'LineStyle', 'none', 'color', 'b', 'MarkerEdgeColor', 'r')
    t = title('Revisit Time Result (L2)', 'FontSize', 12, 'FontWeight', 'bold');
    xlabel('Ground Observation Point
Index', 'FontSize', 11, 'FontWeight', 'bold')
    ylabel('Revisit Time (Hours)', 'FontSize', 11, 'FontWeight', 'bold')
    legend('Min, Max, Mean Revisit Time Data', 'Location', 'southoutside')
    xlim([-1, length(Revisit_Matrix(:,1))+1])

```

